YEN-HSIU, CHEN

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Vision-Language Model Inference / Edge-Cloud Perception Systems / Real-Time Perception-to-Control Integration

SUMMARY

I am a Ph.D. candidate at Yokohama National University (graduating March 2027), making large vision and multimodal models run efficiently on edge devices. My latest method cuts the transmission of large multimodal models (LMMs) by up to 99.73% while keeping up to 99.92% of the original accuracy. I have deployed quantized models on NVIDIA Jetson and Qualcomm NPUs and built real-time perception-to-control systems at CoLab.

EDUCATION

Yokohama National University **Apr 2022 – Mar 2027 (Expected)**

Kanagawa, Japan | M.S. / Ph.D. in Computer Science and Information Engineering

- Graduate School of Environment and Information Sciences (Shirakawa Lab)
- (Fully-funded) Japan-Taiwan Exchange Association Scholarship covering both M.S. and Ph.D. degrees

PUBLICATIONS

C. Yen-Hsiu, et al., "Bandwidth-Constrained Dynamic Split Inference for Large Multimodal Models via Sink-Aware Mixed-Precision Quantization," *IEEE CCNC 2027*.

C. Yen-Hsiu, et al., "Residual Vector Quantization-Based Dynamic Bottleneck for Latency-Accuracy Trade-off in Split Inference," *IEEE Open Journal of the Communications Society (OJCOMS)*. (Under Review)

C. Yen-Hsiu, et al., "Enhancing Latency-Accuracy Tradeoff in Dynamic Split Inference via Vector Quantized Bottleneck," *IEEE GlobeCom Workshops*, 2024, pp. 1–6.

S. Shimizu, . . . , **C. Yen-Hsiu**, et al., "Neural Architecture Search for Improving Latency-Accuracy Trade-off in Split Computing," *IEEE GlobeCom Workshops*, 2022, pp. 1864–1870.

RELEVANT EXPERIENCE

Visiting Researcher **Oct 2026 – Current**

University of Luxembourg, SnT

- Developing split inference for UAV platforms using a real drone dataset, for bandwidth-limited UAV links.
- Contributing to ML-based physical-layer security for UAV-assisted 6G (XTRUST-6G), validating split-inference robustness to channel-induced information loss.

Model Acceleration & Algorithm Intern **Nov 2024 – Feb 2025**

OPPO Inc., Kanagawa

- Built an end-to-end **INT8/FP16 quantization pipeline** for camera and vision models on Snapdragon NPU, covering sensitivity analysis, model conversion, and on-device validation.

AI Research & Development Intern **Aug 2022 – Oct 2024**

CoLab Inc., Kanagawa

- Built a perception-to-control pipeline on industrial cameras and embedded PCs, integrating deep learning with ROS and PLC protocols for low-latency robotic control in a simulated factory.
- Independently built the full ML pipeline, from data collection and training to optimization and internal GUI tools, deployed into factory production. Achieved **97.5% detection accuracy**.

AI Data Analyst Summer Intern **Jul 2021 – Sep 2021**

Lam Research Inc., Taichung, Taiwan

- Trained machine learning models (MARS, Random Forest, GBM) on wafer manufacturing process data to predict outcomes and support troubleshooting, validated together with process and hardware engineers.
- Prepared analysis reports that helped stakeholders follow the analysis and make data-driven decisions.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C++, CUDA, PyTorch
- **Languages:** Mandarin Chinese (Native), English (Proficient), Japanese (Proficient)